

# SMART INDIA HACKATHON 2025



## TITLE PAGE

Problem Statement ID – **SIH26184**

Problem Statement Title – **Development of a Predictive Analytics Framework for Cybercrime Complaints to Forecast Likely Cash Withdrawal Locations in Advance, Enabling Generation of Actionable Intelligence for Timely and Proactive Cybercrime Intervention.**

Theme – **Blockchain & Cybersecurity**

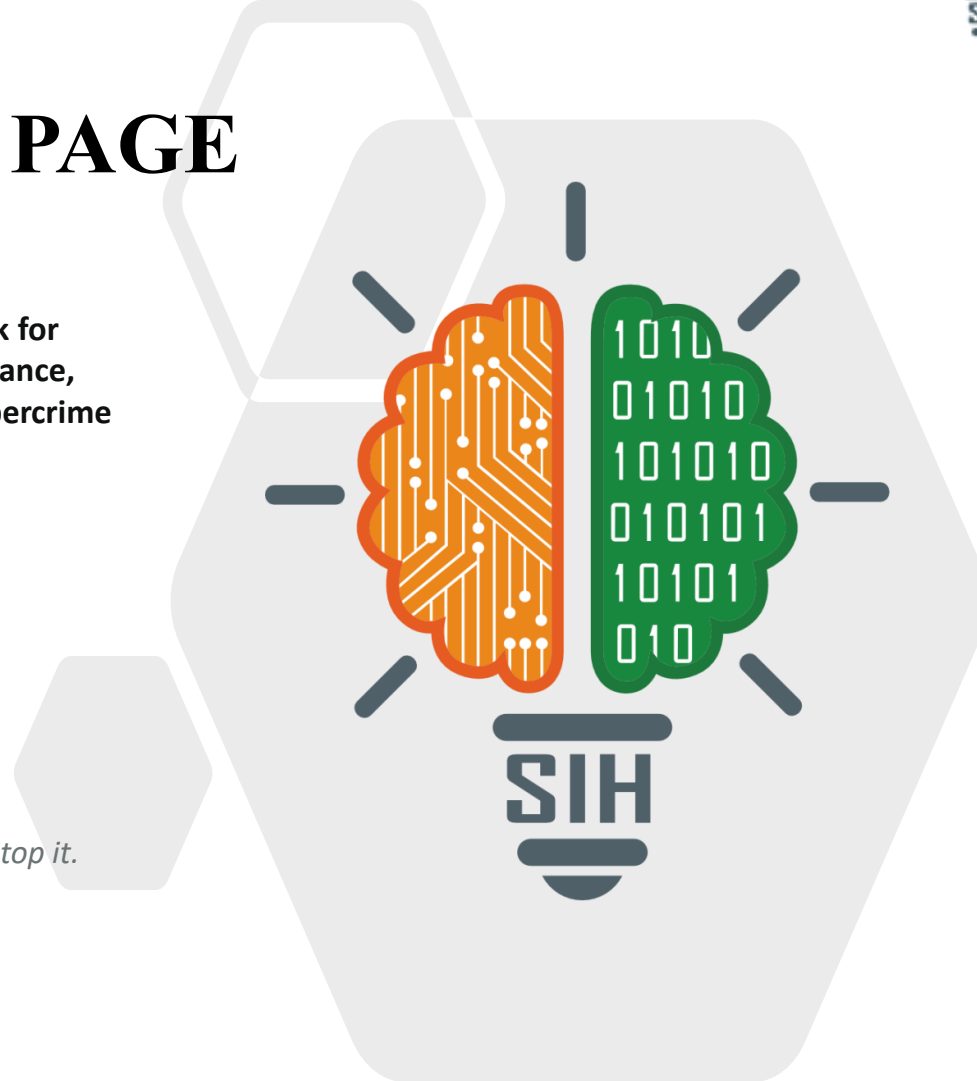
PS Category – **Software**

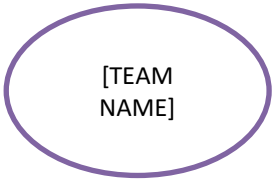
Team ID – **[TEAM ID]**

Team Name – **[TEAM NAME]**

**MuleShield AI**

*Forecasting where stolen money will be withdrawn — while there is still time to stop it.*





[TEAM  
NAME]

# IDEA TITLE



## — Proposed solution

- **The gap:** a 1930 complaint tells you a crime happened. It does not tell you where the cash will leave the system, or when.
- **MuleShield AI** turns that complaint into a ranked, time-bounded search: five ATMs, in order, with a countdown.

## — How it addresses the problem

- **Traces the money.** Builds the transaction graph and follows the chain to the mule account still holding the funds.
- **Scores every account.** A GraphSAGE network reads who-paid-whom, not just per-account behaviour — F1 0.8955.
- **Narrows the ground.** 1,000 ATMs → 25 reachable → 5 worth searching, in 7.5 ms.
- **Starts the clock.** Median 42 minutes of warning before the predicted withdrawal.

### Complaint to intervention — inside the golden hour

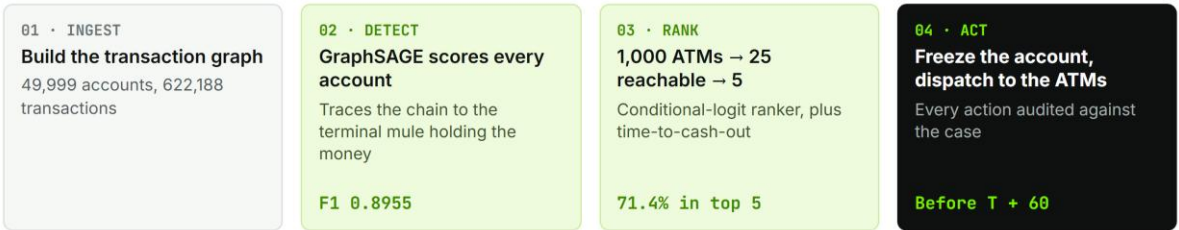
7.5 ms / case

#### ● THE CRIME — THE MONEY IS MOVING



↓ ⌚ INTERVENTION WINDOW — median 42 minutes of warning

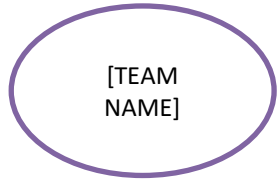
#### ● MULESHIELD AI — TRIGGERED BY THAT COMPLAINT



The system does not name one ATM. It returns the **five worth searching first, in order**, with a countdown — which is what an officer can act on.

## — Innovation and uniqueness

- **Ranked retrieval, not a guess** — a list a field team can cover, never one predicted ATM.
- **Graph-native detection** — the graph is worth +5 F1 points over the best non-graph model.
- **Published honestly** — the console shows its own baselines and limits on screen.
- **Zero new data** — runs on the fields an NCRP complaint already carries.



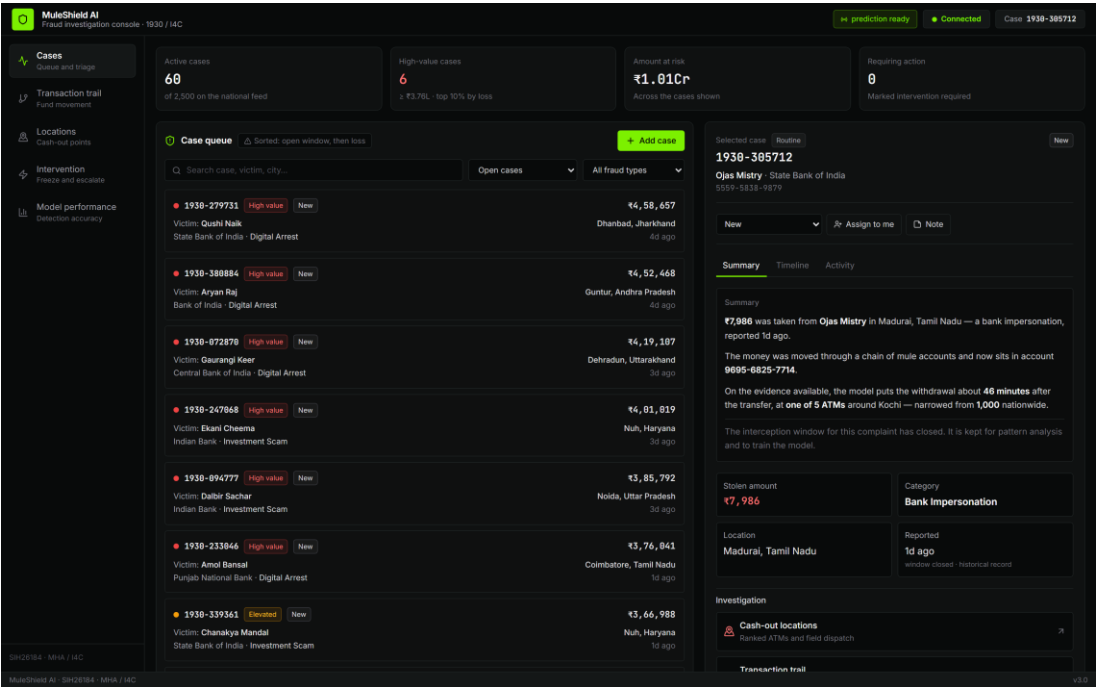
# TECHNICAL APPROACH



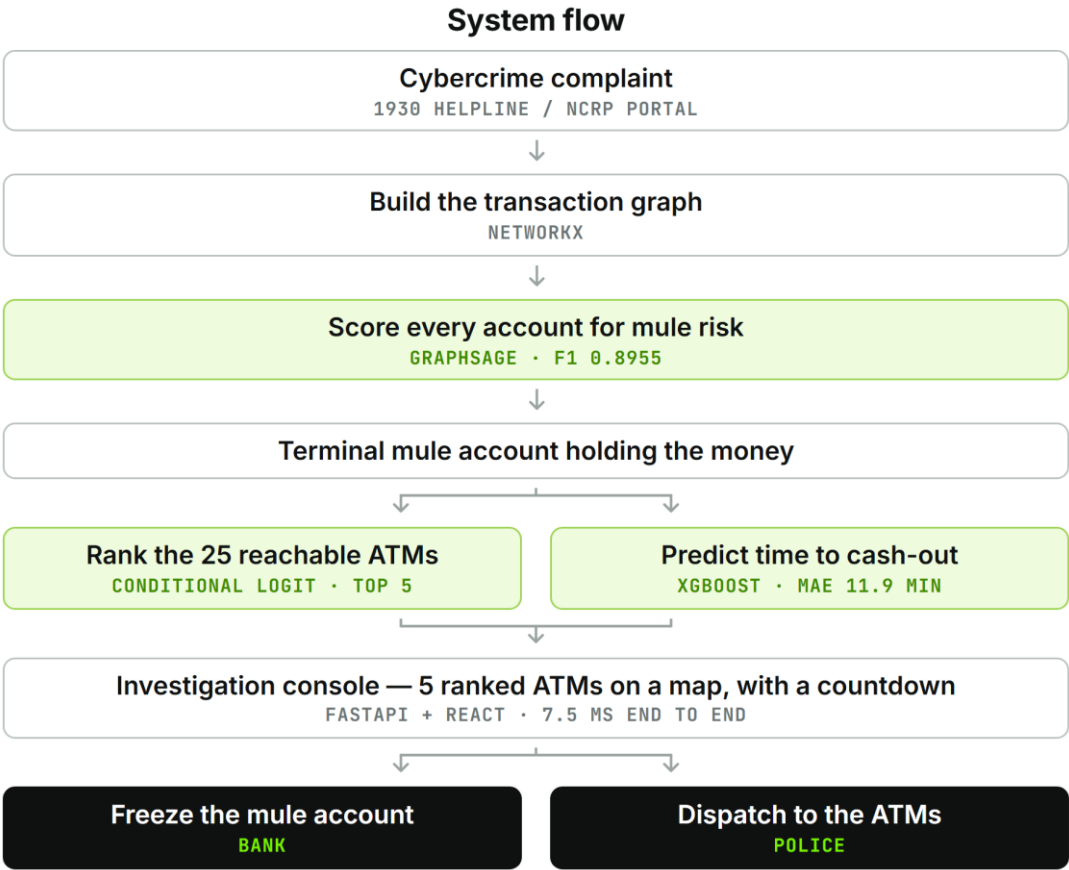
## Technologies used

- ML:** PyTorch · PyTorch Geometric (GraphSAGE) · XGBoost v2 · scikit-learn · NetworkX · pandas / NumPy
- Backend:** Python 3.13 · FastAPI · Uvicorn · Pydantic · WebSockets
- Frontend:** React 19 · Vite · Tailwind · Leaflet (GIS) · React Flow · Recharts
- Quality:** pytest (291 tests) · Playwright · Docker (single container, single process)

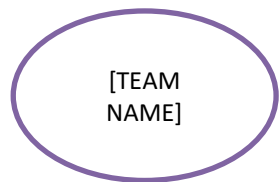
## Working prototype — the case queue



## Methodology and implementation



Every action is written to an append-only **audit trail** against the case.



# FEASIBILITY AND VIABILITY



## — Feasibility

- **Technical:** already built and measured — 291 passing tests, 7.5 ms inference, runs on one CPU container.
- **Data:** consumes the fields NCRP already collects; no new citizen-facing process.
- **Operational:** delivers into the existing 1930 workflow — a queue, a case, an action, an audit trail.
- **Economic:** open-source stack end to end; one container, no GPU at inference.

## — Challenges and how we handle them

- **Leakage inflating results.** Found in our own first build (98.5% claimed). Rebuilt the generator, re-baselined, and added a test that fails if a retracted figure returns.
- **A model that looks confident and is wrong.** Isotonic calibration, verified against held-out labels by a dedicated test.
- **Synthetic training data.** Generated to be hard — 2% label noise, overlapping classes, complaint-level splits. Pipeline retrains on real I4C data unchanged.
- **Acting on a wrong location.** The system returns candidates, never a verdict; the UI and every dispatch alert say so.
- **Single-process state.** Documented limit — case state is in-memory; a database swap is a day's work when multi-user is needed.
- **No inter-bank feed today.** A live complaint carries no transaction trail; production consumes the RBI/NPCI trace-and-report pipeline already being built for CFCFRMS. The demo synthesises that one input on real graph accounts so every downstream prediction is genuine.

## Viability — measured, not asserted

### Model performance matrix

Scored on held-out data the models never saw — complaint-level splits, no tuning on test

| TASK                                                         | OUR MODEL                   | RESULT           | BEST BASELINE                  |
|--------------------------------------------------------------|-----------------------------|------------------|--------------------------------|
| <b>Mule account detection</b><br>F1, 7,500 held-out accounts | GraphSAGE, 2-layer          | <b>0.8955</b>    | 0.8423 random forest, no graph |
| <b>Search-zone containment</b><br>median 8 ATMs of 1,000     | Aggregated probability mass | <b>87.4%</b>     | 78.5% nearest-3 rule           |
| <b>Time-to-cash-out</b><br>mean absolute error               | XGBoost regressor           | <b>11.86 min</b> | 14.98 min mean predictor       |
| <b>Top-5 ATM containment</b><br>618 held-out cash-outs       | Conditional-logit ranker    | <b>71.4%</b>     | 72.2% distance-only            |
| <b>Single-case inference</b><br>complaint to ranked ATMs     | Full pipeline               | <b>7.5 ms</b>    | —                              |

#### DETECTION — READ AS AN ALERT QUEUE

**197**

mules caught

**22**

false alarms

**24**

mules missed

**7,257**

correctly cleared

Of 219 accounts flagged, 197 are mules — an analyst opens about 10 false positives per hundred.

#### HOW OFTEN THE CASH-OUT IS IN THE TOP K

|              |              |
|--------------|--------------|
| Top 1        | 26.5%        |
| Top 3        | 56.1%        |
| <b>Top 5</b> | <b>71.4%</b> |
| Top 10       | 91.8%        |

Five is the shipped operating point — the largest list a field team can actually cover at once.

**Stated openly:** at K=5 the ranker does not beat sorting by distance (71.4% vs 72.2%), and the gap is not statistically significant at any K. Top-1 sits on the theoretical ceiling for this data. Our defensible claim is the narrowing — **1,000 ATMs to 5** — plus the mule account to freeze and the clock, which a distance rule gives you neither of.

# IMPACT AND BENEFITS



## — Who it affects

- **Victims of cyber-financial fraud** — 1930 / NCRP callers, nationwide.
- **Cyber-crime police** acting inside the golden hour.
- **I4C and MHA** — pattern intelligence across cases, not one complaint at a time.
- **Banks** receiving a specific account to freeze.

## — Benefits

- **Social:** recovery becomes possible for ordinary complainants, not only large-value cases.
- **Economic:** cash stopped before withdrawal is money recovered; narrower freezes mean fewer legitimate accounts locked.
- **Operational:** finite patrol capacity aimed at 5 locations instead of a district.
- **Governance:** every freeze and dispatch attributed, timestamped and auditable.
- **Strategic:** recurring cash-out ATMs surface across cases — the mule network, not just the incident.

*Impact figures are held-out measurements, not projections.*

## What changes for the officer on the case

### TODAY

**A complaint, a bank freeze request, and an hour of guessing**

The money is somewhere among 1,000 ATMs. No ranking, no clock, no way to know which mule account still holds it.

### WITH MULESHIELD AI

**Five ranked ATMs, the account to freeze, and time on the clock**

Delivered in 7.5 ms from the complaint that already exists on the NCRP portal.

**1,000**

ATMs nationwide



**25**

reachable from the mule



**5**

to search, in order

**99.5% less ground to cover**, with the actual cash-out among those five **71.4%** of the time — and a median **42 minutes** of warning to get there.

### Citizen

Money stopped while it is still recoverable, instead of a complaint number and a wait. Every 1930 caller is triaged by how much time is left, not by who called first.

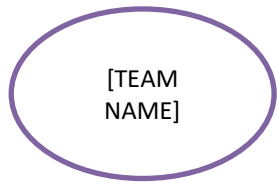
### Police & I4C

Finite patrol capacity aimed at five locations instead of a district. Prior-incident counts surface the ATMs that recur across cases — the network, not just the case.

### Banks

A ranked, calibrated mule account to freeze rather than a blanket hold — fewer legitimate customers locked out, and an audit trail behind every action taken.

Deploys as a single container against data the NCRP portal already collects — **no new reporting burden on the citizen, and no change to how a complaint is filed.**



# RESEARCH AND REFERENCES



## — Problem context

- **I4C, MHA** — Indian Cyber Crime Coordination Centre; 1930 helpline and the NCRP complaint pipeline this system consumes.  
<https://i4c.mha.gov.in/> · <https://cybercrime.gov.in/>
- **RBI** — Master Directions on Fraud Risk Management and mule-account controls in commercial banks.  
[https://www.rbi.org.in/Scripts/BS\\_ViewMasDirections.aspx](https://www.rbi.org.in/Scripts/BS_ViewMasDirections.aspx)
- **NCRB** — Crime in India: cyber-fraud incidence and disposal statistics used to frame the golden-hour problem.  
<https://www.ncrb.gov.in/crime-in-india.html>

## — Methods

- **Hamilton, Ying & Leskovec (2017)** — Inductive Representation Learning on Large Graphs (GraphSAGE). The detection model.  
<https://arxiv.org/abs/1706.02216>

## — Methods (continued)

- **McFadden (1974)** — Conditional Logit Analysis of Qualitative Choice Behavior. The discrete-choice basis of the ATM ranker.
- **Chen & Guestrin (2016)** — XGBoost: A Scalable Tree Boosting System. The countdown regressor.  
<https://arxiv.org/abs/1603.02754>
- **Zadrozny & Elkan (2002)** — Transforming classifier scores into accurate multiclass probability estimates. Isotonic calibration.
- **Weber et al. (2019)** — Anti-Money Laundering in Bitcoin: GCNs for financial forensics. Precedent for GNNs on laundering graphs.  
<https://arxiv.org/abs/1908.02591>

## — Our own work

- **Source, evaluation and write-up** — full implementation, 291 tests, the leakage audit, and the architecture in long form.  
<https://github.com/hotshot0104/SIH2026>  
<https://dev.to/sameer0104/muleshield-ai-intercepting-cybercrime-cash-outs-before-the-trail-goes-cold-232g>

### What the three models optimise

**1 Mule detection**  
GRAPHSAGE · 2 LAYERS · 64-DIM

$$h_v^{(k)} = \text{ReLU}(W^{(k)} \cdot [h_v^{(k-1)} \parallel \text{mean}_{u \in N(v)} h_u^{(k-1)}])$$

$$p(\text{mule}) = \sigma(w \cdot h_v^{(2)} + b)$$

Class-weighted cross-entropy, then isotonic calibration so the probability means what it says.

**2 Cash-out location**  
CONDITIONAL LOGIT · 25 CANDIDATES

$$P(\text{atm}_i \mid C) = e^{w \cdot \log x_i} / \sum_{j \in C} e^{w \cdot \log x_j}$$

$$\max_w \sum_{\text{groups}} \log P(\text{chosen} \mid C)$$

Choice is multiplicative (proximity × risk × affinity × habit), so logs make it linear.

**3 Time to cash-out**  
GRADIENT BOOSTING · XGBOOST

$$\hat{y} = \sum_{m=1}^M f_m(x), \quad f_m \in F$$

$$\min \sum (\gamma - \hat{y})^2 + \sum_m \Omega(f_m)$$

Predicts minutes to withdrawal; q05 / q95 models add a band, not a false point estimate.